DESeq2 moderated dispersion and fold-change estimation

A paper talk on negative-binomial RNA-seq inference

Research Presentation Program

Department of Statistics & Data Science

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DESeq2 starts from a negative-binomial GLM

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i), \quad \mu_{ij} = s_j q_{ij}, \quad \log_2 q_{ij} = x_{j.}\beta_i$$

The GLM frame lets DESeq2 estimate expression effects while modeling mean-dependent count variance.

Model components	**Interpretation**
• negative binomial counts • size factors • dispersion alpha_i	The model separates sequencing-depth normalization, expected expression, and gene-specific dispersion.

Love, Huber, and Anders 2014, Methods.

Fold-change shrinkage stabilizes weakly informed effects

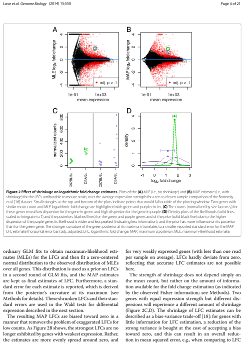

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Figure 2 Effect of shrinkage on logarithmic fold change estimates. Plots of the **(A)** MLE (i.e., no shrinkage) and **(B)** MAP estimate (i.e., with shrinkage) for the LFCs attributable to mouse strain, over the average expression strength for a ten vs eleven sample comparison of the Bottomly *et al.* [16] dataset. Small triangles at the top and bottom of the plots indicate points that would fall outside of the plotting window. Two genes with similar mean count and MLE logarithmic fold change are highlighted with green and purple circles. **(C)** The counts (normalized by size factors s_j) for these genes reveal low dispersion for the gene in green and high dispersion for the gene in purple. **(D)** Density plots of the likelihoods (solid lines, scaled to integrate to 1) and the posteriors (dashed lines) for the green and purple genes and of the prior (solid black line): due to the higher dispersion of the purple gene, its likelihood is wider and less peaked (indicating less information), and the prior has more influence on its posterior than for the green gene. The stronger curvature of the green posterior at its maximum translates to a smaller reported standard error for the MAP LFC estimate (horizontal error bar). adj., adjusted; LFC, logarithmic fold change; MAP, maximum *a posteriori*; MLE, maximum-likelihood estimate.

ordinary GLM fits to obtain maximum-likelihood estimates (MLEs) for the LFCs and then fit a zero-centered normal distribution to the observed distribution of MLEs over all genes. This distribution is used as a prior on LFCs in a second round of GLM fits, and the MAP estimates are kept as final estimates of LFC. Furthermore, a standard error for each estimate is reported, which is derived from the posterior's curvature at its maximum (see Methods for details). These shrunken LFCs and their standard errors are used in the Wald tests for differential expression described in the next section.

The resulting MAP LFCs are biased toward zero in a manner that removes the problem of exaggerated LFCs for low counts. As Figure 2B shows, the strongest LFCs are no longer exhibited by genes with weakest expression. Rather, the estimates are more evenly spread around zero, and for very weakly expressed genes (with less than one read per sample on average), LFCs hardly deviate from zero, reflecting that accurate LFC estimates are not possible here.

The strength of shrinkage does not depend simply on the mean count, but rather on the amount of information available for the fold change estimation (as indicated by the observed Fisher information; see Methods). Two genes with equal expression strength but different dispersions will experience a different amount of shrinkage (Figure 2C,D). The shrinkage of LFC estimates can be described as a bias-variance trade-off [18]: for genes with little information for LFC estimation, a reduction of the strong variance is bought at the cost of accepting a bias toward zero, and this can result in an overall reduction in mean squared error, e.g., when comparing to LFC

Shrinkage depends on information: low counts, high dispersion, or few degrees of freedom pull LFC estimates more strongly toward zero.

Figure 2 page, Love et al. 2014.

Dispersion shrinkage borrows strength across genes

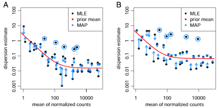


Figure 1 Shrinkage estimation of dispersion. Plot of dispersion estimates over the average expression strength **(A)** for the Bottomly *et al.* [16] dataset with six samples across two groups and **(B)** for five samples from the Pickrell *et al.* [17] dataset, fitting only an intercept term. First, gene-wise MLEs are obtained using only the respective gene's data (black dots). Then, a curve (red) is fit to the MLEs to capture the overall trend of dispersion-mean dependence. This fit is used as a prior mean for a second estimation round, which results in the final MAP estimates of dispersion (arrow heads). This can be understood as a shrinkage (along the blue arrows) of the noisy gene-wise estimates toward the consensus represented by the red line. The black points circled in blue are detected as dispersion outliers and not shrunk toward the prior (shrinkage would follow the dotted line). For clarity, only a subset of genes is shown, which is enriched for dispersion outliers. Additional file 1: Figure S1 displays the same data but with dispersions of all genes shown. MAP, maximum *a posteriori*; MLE, maximum-likelihood estimate.

variation to the extent that the data provide this information, while the fitted curve aids estimation and testing in less information-rich settings.

Our approach is similar to the one used by *DSS* [6], in that both methods sequentially estimate a prior distribution for the true dispersion values around the fit, and then provide the maximum *a posteriori* (MAP) as the final estimate. It differs from the previous implementation of *DESeq*, which used the maximum of the fitted curve and the gene-wise dispersion estimate as the final estimate and tended to overestimate the dispersions (Additional file 1: Figure S2). The approach of *DESeq2* differs from that of *edgeR* [3], as *DESeq2* estimates the width of the prior distribution from the data and therefore automatically controls the amount of shrinkage based on the observed properties of the data. In contrast, the default steps in *edgeR* require a user-adjustable parameter, the *prior degrees of freedom*, which weighs the contribution of the individual gene estimate and *edgeR*'s dispersion fit.

Note that in Figure 1 a number of genes with gene-wise dispersion estimates below the curve have their final estimates raised substantially. The shrinkage procedure thereby helps avoid potential false positives, which can result from underestimates of dispersion. If, on the other hand, an individual gene's dispersion is far above the distribution of the gene-wise dispersion estimates of other genes, then the shrinkage would lead to a greatly reduced final estimate of dispersion. We reasoned that in many cases, the reason for extraordinarily high dispersion of a gene is that it does not obey our modeling assumptions; some genes may show much higher variability than others for biological or technical reasons, even though they have the same average expression levels. In these cases, inference based on the shrunken dispersion estimates could lead to undesirable false positive calls. *DESeq2* handles these cases by using the gene-wise estimate instead of the shrunken estimate when the former is more than 2 residual standard deviations above the curve.

Empirical Bayes shrinkage for fold-change estimation

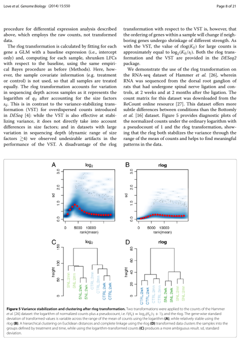
A common difficulty in the analysis of HTS data is the strong variance of LFC estimates for genes with low read count. We demonstrate this issue using the dataset by Bottomly *et al.* [16]. As visualized in Figure 2A, weakly expressed genes seem to show much stronger differences between the compared mouse strains than strongly expressed genes. This phenomenon, seen in most HTS datasets, is a direct consequence of dealing with *count* data, in which ratios are inherently noisier when counts are low. This heteroskedasticity (variance of LFCs depending on mean count) complicates downstream analysis and data interpretation, as it makes effect sizes difficult to compare across the dynamic range of the data.

DESeq2 overcomes this issue by shrinking LFC estimates toward zero in a manner such that shrinkage is stronger when the available information for a gene is low, which may be because counts are low, dispersion is high or there are few degrees of freedom. We again employ an empirical Bayes procedure: we first perform

Figure 1 page, Love et al. 2014, CC BY.

The blue arrow shrinkage in Figure 1 is the core mechanism: noisy estimates move toward a fitted consensus unless treated as outliers.

Benchmark gains are empirical, not a license to ignore design



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procedure for differential expression analysis described above, which employs the raw counts, not transformed data.

The rlog transformation is calculated by fitting for each gene a GLM with a baseline expression (i.e., intercept only) and, computing for each sample, shrunken LFCs with respect to the baseline, using the same empirical Bayes procedure as before (Methods). Here, however, the sample covariate information (e.g. treatment or control) is not used, so that all samples are treated equally. The rlog transformation accounts for variation in sequencing depth across samples as it represents the logarithm of q_{ij} after accounting for the size factors s_{ij}. This is in contrast to the variance-stabilizing transformation (VST) for overdispersed counts introduced in *DESeq* [4]: while the VST is also effective at stabilizing variance, it does not directly take into account differences in size factors; and in datasets with large variation in sequencing depth (dynamic range of size factors $\gtrsim 4$) we observed undesirable artifacts in the performance of the VST. A disadvantage of the rlog transformation with respect to the VST is, however, that the ordering of genes within a sample will change if neighboring genes undergo shrinkage of different strength. As with the VST, the value of $\mathrm{rlog}(K_{ij})$ for large counts is approximately equal to $\log_2(K_{ij}/s_j)$. Both the rlog transformation and the VST are provided in the *DESeq2* package.

We demonstrate the use of the rlog transformation on the RNA-seq dataset of Hammer *et al.* [26], wherein RNA was sequenced from the dorsal root ganglion of rats that had undergone spinal nerve ligation and controls, at 2 weeks and at 2 months after the ligation. The count matrix for this dataset was downloaded from the ReCount online resource [27]. This dataset offers more subtle differences between conditions than the Bottomly *et al.* [16] dataset. Figure 5 provides diagnostic plots of the normalized counts under the ordinary logarithm with a pseudocount of 1 and the rlog transformation, showing that the rlog both stabilizes the variance through the range of the mean of counts and helps to find meaningful patterns in the data.

Figure 5 Variance stabilization and clustering after rlog transformation. Two transformations were applied to the counts of the Hammer *et al.* [26] dataset: the logarithm of normalized counts plus a pseudocount, i.e. $f(K_{ij}) = \log_2(K_{ij}/s_j + 1)$, and the rlog. The gene-wise standard deviation of transformed values is variable across the range of the mean of counts using the logarithm **(A)**, while relatively stable using the rlog **(B)**. A hierarchical clustering on Euclidean distances and complete linkage using the rlog **(D)** transformed data clusters the samples into the groups defined by treatment and time, while using the logarithm-transformed counts **(C)** produces a more ambiguous result. sd, standard deviation.

Figure 6 supports comparative performance claims, but the paper still frames testing around design, FDR, and effect-size thresholds.

Figure 6 page, Love et al. 2014.

The durable idea is moderated estimation, not more filters

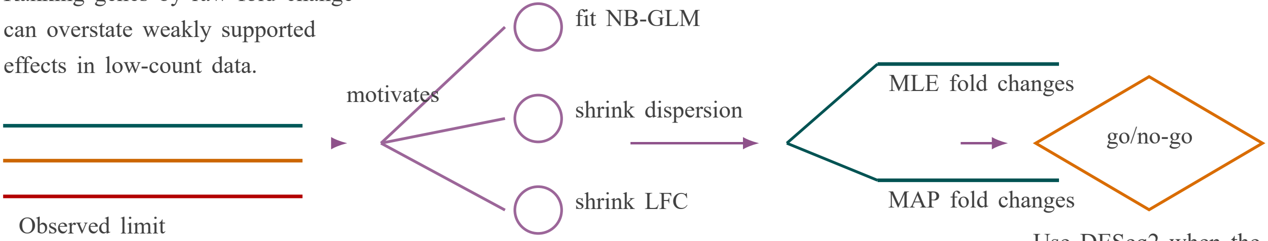
Failure evidence

Ranking genes by raw fold change can overstate weakly supported effects in low-count data.

Manipulate sampling

Comparator arms

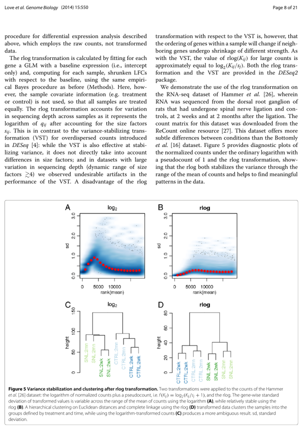
Decision rule



Use DESeq2 when the scientific question benefits from moderated dispersion and effect-size estimates with transparent design assumptions.

The paper-specific close is an estimation contract: regularize noisy gene-wise quantities before ranking or testing.

The comparison object is sensitivity versus precision



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This comparison slot remains a DESeq2 benchmark page: it is about differential-expression algorithms, not a generic omics pipeline.

Algorithm-comparison evidence from Figure 6.